

KAPREKAR CONTEST – FINAL – SUB-JUNIOR

AMTI – OCTOBER, 2009.



Instructions:

1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Fill in FACE SLIP and your rough working should be in the answer book itself at the allotted place.
5. Maximum time allowed is THREE hours.

1. Find the number of distinct remainders when 2009 is divided by natural numbers. Note that the remainder r is such that $0 \leq r < 2009$.

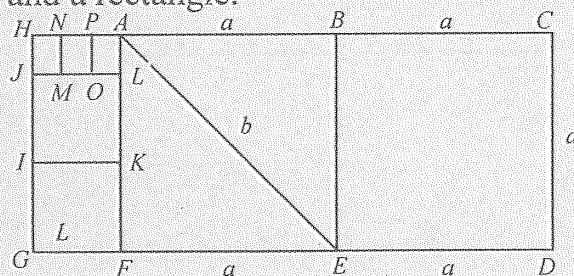
2. Find the sum of all the positive integers less than 2009 and relatively prime to 2009.

3. There are two neighboring townships A and B . In township A there are 500 children. When there was spread of an epidemic, in township B 40% more students were free from the attack of the epidemic compared to township A . If the number of students free from the disease is the same in both the townships, find the student population in township B .

4. If a, b, n are natural numbers and $n \neq 1$ arrange the following in ascending order of magnitude.

$$\frac{na}{(n+1)b}, \quad \frac{(n+1)a}{nb}, \quad \frac{na}{(n-1)b}, \quad \frac{(n-1)a}{nb}$$

5. In the diagram given below $ABEF$, $BCDE$ are squares of side ' a ' units and EG (the extension of EF) is ' b ' units. In the figure, the rectangles are successively dissected into pairs of squares and a rectangle.

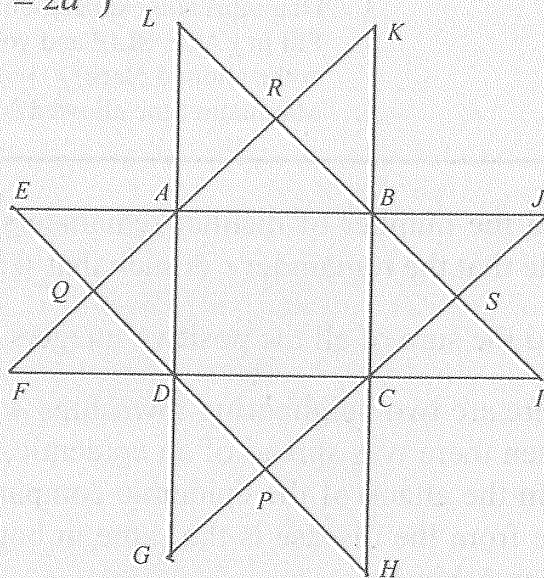


The first four stages of this process is shown in the figure. The dimensions of the rectangles in these stages are $(a+b, a)$, $(a, b-a)$, $(b-a, 3a-2b)$ and $(3a-2b, 5b-7a)$ respectively. The ratio of the length to the breadth in each case are respectively.

$$\frac{a+b}{a}, \quad \frac{a}{b-a}, \quad \frac{b-a}{3a-2b}, \quad \frac{3a-2b}{5b-7a}$$

Find the ratio of the length to breadth for 5 more stages and show that all the nine ratios are each equal to $\frac{a+b}{a}$ (Note that $b^2 = 2a^2$)

6. In the adjoining figure $ABCD$ is a square of unit length. The side AB , BC , CD and DA are produced on both the directions so that the extended segments also have lengths 1 unit. Show that the point triples (L, B, I) , (K, A, F) , (J, C, G) and (F, D, I) are collinear and $PQRS$ is a square.



7. a, b, c are distinct digits. Find all (a, b, c) such that the three digit numbers abc and cba are both divisible by 7.
8. x, y, z are positive numbers such that

$$x = \frac{2y}{1+y}, \quad y = \frac{2z}{1+z}, \quad z = \frac{2x}{1+x}$$

Prove that $x = y = z$
